

Chapter - 1: Electric Charges and Fields

~ Complete Handwritten Physics Lecture Notes ~

Electric Charge:

A property of substance which arises due to either loss or gain of electron by an atom.

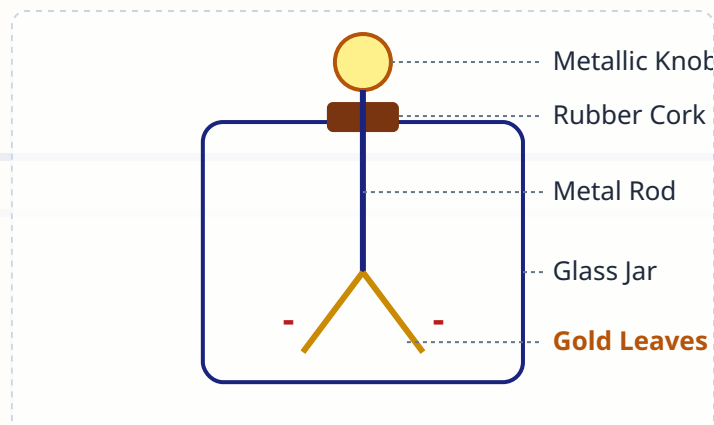
- S.I. Unit: Coulomb (C)
- Elementary Charge values:
 - Charge on 1 electron (e^-) = $-1.6 \times 10^{-19} \text{ C}$
 - Charge on 1 proton (p^+) = $+1.6 \times 10^{-19} \text{ C}$
- There are only 2 kinds of electric charges: Positive & Negative.
- Fundamental Law of Electrostatics: Like charges repel each other and unlike charges attract each other.

Electrostatic Induction:

The process in which redistribution of charge takes place within a body when an electrically charged body is brought near to it.

~ Gold-leaf Electroscope:

Device which is used to detect the presence of charge on a body by the principle of electrostatic induction.



[Fig: Gold Leaf Electroscope Setup]

Basic Property of Electric Charge:

1) Additivity of electric charge:

It means that the total charge of a system is the algebraic sum of all the individual charges located at different points inside the system.

2) Conservation of charge:

- Total charge of an isolated system remains constant.
- The electric charges can neither be created nor be destroyed, they can only be transferred from one body to another.

3) Quantisation of electric charge:

It means that electric charge cannot vary continuously to have any arbitrary value, but it can change discontinuously to take any one of only a discrete set of values.

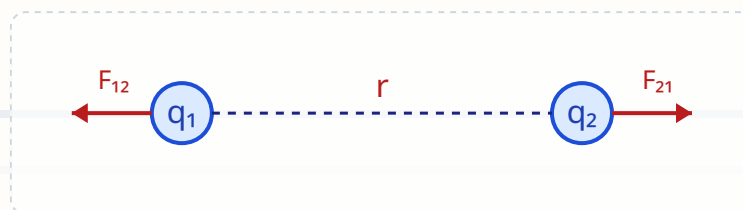
The experimental fact that electric charges occur in discrete amounts instead of continuous amounts is called quantisation of electric charge.

$$Q = \pm n e$$

The total charge (Q) of a body is always an integral multiple of basic quantum of charge.

Coulomb's Law of Electrostatics:

Electrostatic force between 2 point charges is:



[Fig: Electrostatic Force Between Charges]

- i) $F \propto q_1 \times q_2$
- ii) $F \propto 1 / r^2$

So, $F \propto (q_1 q_2 / r^2)$

$$\mathcal{F} = k \cdot (q_1 q_2 / r^2)$$

k = electrostatic force constant (medium dependent)

For free space, $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

Also, $k = 1 / (4\pi \epsilon_0)$ ϵ_0 = Permittivity of free space = $8.855 \times 10^{-12} \text{ N}^{-1} \text{ m}^{-2} \text{ C}^2$

$$\mathcal{F} = (1 / 4\pi \epsilon_0) \cdot (q_1 q_2 / r^2)$$

Note: Relative permittivity = Dielectric constant = K

K = (Permittivity of medium) / (Permittivity of free space)

Electric Field ($\mathcal{E}$):

Region around a charge up to which it can apply electrostatic force on other charges.

OR Electrostatic force per unit test charge:

$$\mathcal{E} = \mathcal{F} / q$$

* *Electric Field for a point charge:*

As, $\mathcal{F} = k (q q_0 / r^2)$

Also, $\mathcal{E} = \mathcal{F} / q_0 = [k (q q_0 / r^2)] / q_0$

$$\mathcal{E} = k q / r^2 = (1 / 4\pi \epsilon_0) \cdot (q / r^2)$$

Electric Dipole:

System of two equal and opposite charges separated by a distance ($2a$).

* *Electric Dipole Moment (p):*

Product of either charge with distance between the charges:

$$p = q \times 2a$$

It is a vector quantity (-ve to +ve charge).

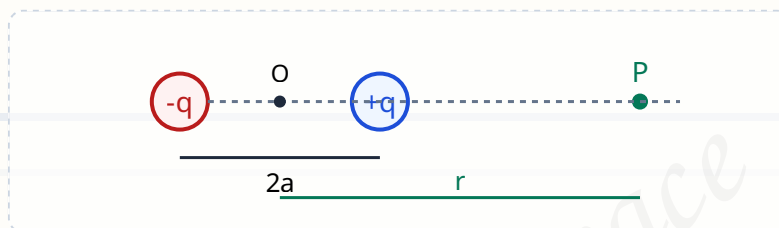
Dipole Field:

* Electric field at axial point of dipole:

$$E_{+q} = Kq / (r + a)^2$$

$$E_{-q} = -Kq / (r - a)^2$$

$$E_{net} = E_{+q} + E_{-q} = Kq [1/(r + a)^2 - 1/(r - a)^2]$$



[Fig: Axial Point of Dipole]

$$E_{ax} = -2 Kp r / (r^2 - a^2)^2$$

For short dipole ($a \ll r$), neglecting a^2 :

$$E_{ax} = -2 Kp / r^3$$

* Electric field at equatorial point of dipole:

$$E_{eq} = E_{+q} \cos\theta + E_{-q} \cos\theta = 2 E_{+q} \cos\theta$$

$$E_{eq} = 2 Kq a / [(r^2 + a^2)^{3/2}]$$

$$E_{eq} = Kp / (r^2 + a^2)^{3/2}$$

For short dipole ($a \ll r$), neglecting a :

$$E_{eq} = Kp / r^3$$

* Torque on a dipole in a uniform electric field:

$$F_{+q} = E q$$

$$F_{-q} = -E q$$

$$F_{net} = F_{+q} + F_{-q} = E q - E q = 0$$

Now, $\tau = F \times \perp \text{ dist} = q E \times 2a \sin\theta$

$$\tau = p E \sin\theta \text{ \& implies; } \vec{\tau} = \vec{p} \times \vec{E}$$

Special Cases:

- $\theta = 0^\circ$ \& implies; $\tau = 0$ (Stable Eq.^b)
- $\theta = 90^\circ$ \& implies; $\tau = p E = \max$
- $\theta = 180^\circ$ \& implies; $\tau = 0$ (Unstable Eq.^b)

Electric Field Lines:

An electric line of force may be defined as the curve along which a small positive charge would tend to move when free to do so in an electric field, and the tangent to which at any point gives the direction of electric field at that point.

* *Properties of electric lines of force:*

1. The lines of force are continuous smooth curves without any breaks.
2. The lines of force start at positive charges and end at negative charges – they cannot form closed loops.
3. The tangent to the line of force at any point gives the direction of electric field at that point.
4. No two lines of force can cross each other. Reason: If they intersect, then there will be two tangents at that point of intersection and hence 2 directions of electric field at a point, which is not possible.
5. The lines of force are always normal to the surface of conductor on which the charges are in equilibrium.
6. Relative closeness of lines of force gives measure of strength of electric field:
i. Close together in strong field ii. Far apart in weak field iii. Parallel in uniform field.
7. The lines of force do not pass through a conductor because electric field inside a charged conductor is zero (Electrostatic Shielding).

Area Vector:

For closed body, area vector is always $\perp$ outwards.

Electric Flux (Φ_E):

No. of electric field lines passing through an area $\perp$ arly.

$$\Phi_E = E S \cos\theta = \vec{E} \cdot \vec{S}$$

$$d\Phi_E = \vec{E} \cdot d\vec{S} \quad \text{S.I. unit} = \mathcal{N} m^2 C^{-1}$$

Gauss's Theorem:

Gauss theorem states that the total flux through a closed surface is $1/\epsilon_0$ times the net charge enclosed by closed surface.

$$\Phi_E = q / \epsilon_0$$

Proof:

$$\Phi_E = \oint \vec{E} \cdot d\vec{S} = \oint E dS \cos 0^\circ = E \oint dS$$

$$\Phi_E = (kq/r^2) \cdot 4\pi r^2 = [1/(4\pi \epsilon_0) \cdot (q/r^2)] \times 4\pi r^2$$

$$\Phi_E = q / \epsilon_0$$

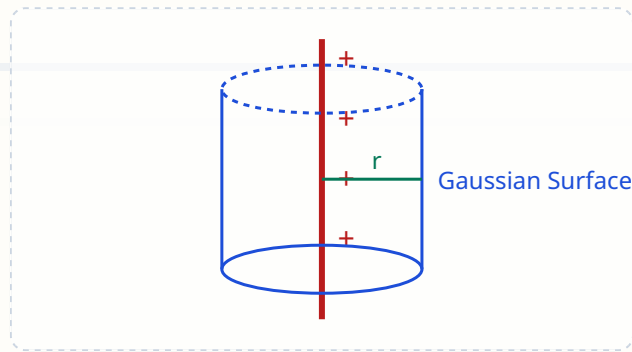
Applications of Gauss's Law:

** Electric field due to long infinite wire:*

$$\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0$$

$$\int E dS_1 \cos 90^\circ + \int E dS_2 \cos 0^\circ + \int E dS_3 \cos 90^\circ = q / \epsilon_0$$

$$\int E dS_2 = q / \epsilon_0 \text{ \&implies; } E \times 2\pi r l = q / \epsilon_0$$



[Fig: Cylindrical Gaussian Surface]

$$E = \lambda / (2\pi r \epsilon_0) \quad (\because \lambda = q / l)$$

Note: If we put a charge on a vertex, then by assuming this point as centre, it will form 8 more cubes.

** Electric field due to uniformly charged infinite plane sheet:*

$$\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0$$

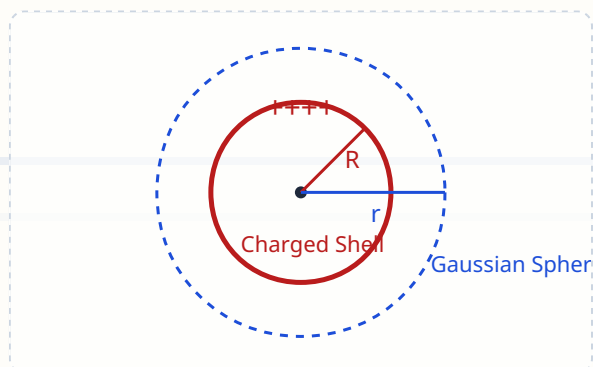
$$\int E dS_1 \cos 0^\circ + \int E dS_2 \cos 90^\circ + \int E dS_3 \cos 0^\circ = q / \epsilon_0$$

$$2 \int E dS = q / \epsilon_0 \text{ \&implies; } 2 E A = q / \epsilon_0$$

$$E = \sigma / (2 \epsilon_0) \quad (\text{Constant})$$

** Electric field due to uniformly thin spherical shell:*

$$i) \text{ Inside: } \oint \vec{E} \cdot d\vec{S} = q / \epsilon_0 = 0 \text{ \&implies; } q = 0$$



[Fig: Spherical Shell & Outer Gaussian Sphere]

$$ii) \text{ Outside: } \oint \vec{E} \cdot d\vec{S} = q / \epsilon_0 \text{ \&implies; } E \int dS = q / \epsilon_0$$

$$E = q / (4\pi \epsilon_0 r^2)$$

iii) On the surface:

$$E = q / (4\pi \epsilon_0 R^2)$$

* Coulomb's law from Gauss's law:

Acc. to Gauss law: $\Phi = q / \epsilon_0$

$$\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0 \text{ implies; } E \times 4\pi R^2 = q / \epsilon_0$$

$$E = q / (4\pi \epsilon_0 R^2)$$

$$\therefore F = E q_0 \text{ implies; } F = [q / (4\pi \epsilon_0 R^2)] \times q_0$$

$$F = K q q_0 / R^2$$